



Meta-analysis in public administration research: An interdisciplinary perspective review in the age of open science

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Abstract

This study combines content analysis and systematic literature review to examine the landscape of meta-analyses in public administration. Content analysis of 107 articles shows that research spans both micro- and macro-level topics, with particular emphasis on public policy, governance, and service delivery. We propose a structured framework for conducting meta-analyses and critically review 34 studies from five leading journals published over the past decade (2015–2024). Building on these findings, we offer four interdisciplinary recommendations to enhance methodological rigor, strengthen theoretical integration, and foster cumulative knowledge development in the era of open science. By highlighting the potential of meta-analysis to reconcile divergent findings, improve cross-national comparability, and inform evidence-based policy, this study consolidates its role as a foundational tool in public administration scholarship.

Points for practitioners:

- Meta-analysis equips public administrators and policymakers with a rigorous tool to synthesize empirical findings, reconcile inconsistent results, and compare policy effects across diverse institutional and national contexts.
- Meta-analyses in public administration address both macro-level issues (e.g. public policy, service delivery, climate governance) and micro-level concerns (e.g. trust,

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citizen satisfaction, employee motivation), highlighting their relevance for evidence-informed decision making.

- Researchers should follow established protocols while embracing open science practices—transparency, reproducibility, and contextual sensitivity—to enhance the credibility, comparability, and practical relevance of meta-analytic findings.

Keywords

Comparative public administration, literature review, meta-analysis, open science, public administration

Introduction

Literature reviews serve as a mechanism to integrate findings from different yet related studies, synthesizing prior studies to strengthen the foundation of knowledge (Paul and Criado, 2020). Nevertheless, qualitative literature reviews inherently possess subjectivity as they rely on authors' perspectives to determine which studies to incorporate and how best to synthesize them (Borenstein et al., 2021). Researchers advocate for a more systematic approach in conducting review-type articles (DeSimone et al., 2019; Light and Smith, 1971), which has led to the flourishing development of meta-analysis. Meta-analysis is "*the analysis of analyses*" emphasis (Glass, 1976), regarded as the top of what is known as the "evidence pyramid". Unlike narrative reviews that summarize different study findings based on the quantity of statistically significant results, meta-analysis serves as an examination and integration of numerous similar empirical analyses, intending to interpret existing literature on a particular relationship comprehensively (Hur and Abner, 2024; Stanley, 2001; Stanley and Jarrell, 2005). It is a systematic literature review conducted from a quantitative perspective aimed at re-analyzing existing quantitative studies (DeSimone et al., 2019), providing the basics for policies or decisions (Moreau and Gamble, 2022).

Meta-analysis is particularly vital in public administration owing to the field's diverse and often conflicting empirical findings, which stem from varying administrative systems, cultural contexts, and methodological approaches (Chen and Ganapati, 2023; Ding and Riccucci, 2023; Homberg et al., 2015; Wirtz and Daiser, 2016). By systematically synthesizing quantitative studies, meta-analysis provides a robust tool to reconcile these discrepancies, offering generalizable insights that inform evidence-based policy-making and advance comparative public administration (Park, 2020; Parola et al., 2019). Over the past 10 years, meta-analysis has emerged as a central tool for synthesizing empirical evidence, reconciling conflicting findings, and generating robust conclusions that support evidence-based decision making in public policy and management, including comparative public administration research across national and institutional contexts (Ding et al., 2021; George et al., 2024; Zhang et al., 2023).

Existing scholarship has acknowledged the methodological and epistemological relevance of meta-analysis for public administration, highlighted its potential to reconcile divergent empirical findings (Perry, 2012), and explored its application in the field

(Ringquist, 2013). However, existing work can be further improved in three respects. First, most of these writings are primarily methodological introductions or discipline-wide reflections, and they fall short of systematically reviewing how meta-analysis has actually been applied within public administration, especially in the past 10 years. Second, they do not situate meta-analysis in the context of the contemporary open science movement, which imposes new demands for transparency, reproducibility, and interdisciplinary comparison. Third, there has been little attention to the role of meta-analysis in comparative public administration—a subfield in which national contexts and administrative traditions are particularly consequential, yet where systematic quantitative syntheses remain scarce. These gaps underscore the need for the present article, which seeks to provide a more integrated, interdisciplinary, and context-sensitive account of meta-analysis in public administration.

Combining content analysis and systematic literature review, this research makes three key contributions. First, we visually map 107 existing meta-analyses in public administration and develop an analytical framework for future meta-analytic research. Second, we conduct a systematic review of 34 meta-analytic studies published over the past decade in five leading journals, detailing the procedures of meta-analysis and addressing issues of methodological rigor, comprehensiveness, and standardization. Third, we propose interdisciplinary recommendations to strengthen standardization, enhance theoretical robustness, and improve the cross-national comparability of empirical research in public administration. Overall, our study not only advances the theoretical robustness of meta-analytic findings but also improves their credibility and reliability, thereby contributing to the cumulative development of knowledge in comparative public administration in the era of open science.

Methodology

We conducted a systematic search on the Web of Science databases using the following search terms: “*meta-analysis OR meta analysis OR meta-analytic* OR meta analytic**”, restricted to the Web of Science category “*Public Administration*”, and covering the period from the database’s inception to 31 December 2024. Subsequently, we excluded non-English articles, resulting in an initial pool of 245 journal articles, which were then downloaded into EndNote software, including their titles, abstracts, and publication titles for initial screening (Blalock et al., 2023). Referring to the Social Science Citation Index (SSCI) journal list 2023, we identified 202 articles in the *Public Administration* category. After reviewing the title and abstract of each article, 95 articles that did not focus on systematic-review and meta-analysis research were excluded, resulting in 107 articles that aligned with the theme of meta-analysis (see Figure 1).

We then employed Leximancer, a software tool specializing in textual data analysis, particularly in text mining and artificial learning (Smith and Humphreys, 2006). Leximancer stands out for its ability to identify groups of related terms within texts based on their frequency and occurrence patterns. Leximancer offers distinct advantages over conventional content analysis methods (Fan and Lyu, 2021). It is a comprehensive platform designed to analyze text data conceptually, independent of language barriers

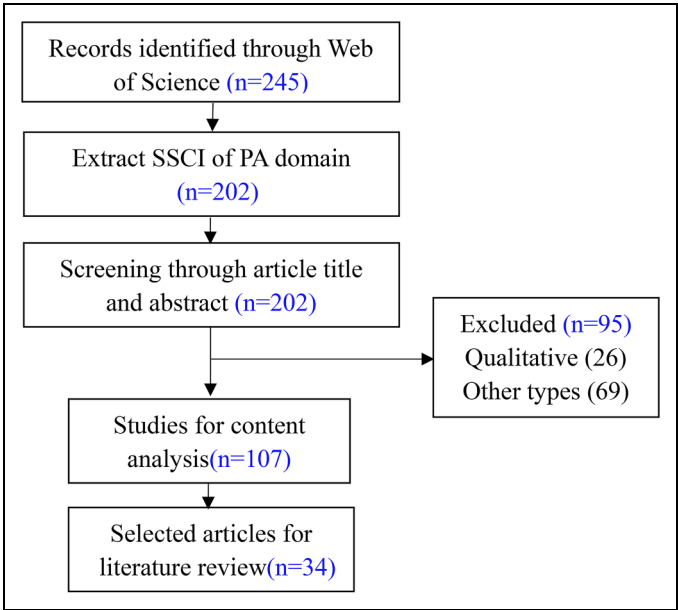


Figure 1. Outline of the literature search process.

(Blalock et al., 2023). Using the cluster map and the concept lists ranked by importance, we delved into the relationships between these concepts to gain a deeper understanding of the semantic structure of the data.

Content analysis

Before proceeding with data analysis, we conducted preprocessing steps in accordance with Leximancer’s standard exclusion criteria, which involved filtering out non-lexical and weak semantic information, including common words such as “of”, “and”, and “not”. Additionally, frequently occurring general terms found in the abstracts, such as “article”, “analysis”, “use”, “study”, “research”, and “results” were removed as they contributed little to the conceptual maps. Furthermore, singular and plural forms of words were merged into their plural forms to ensure consistency (e.g. “policy” and “policies”).

Moreover, to maintain the comprehensiveness of the concept map, we set the visibility of concepts to 100%, ensuring that all concepts were displayed on the map while limiting the size of themes to 50%. Subsequently, leveraging the cluster map (indicating concept size and connections) and the concept lists ordered by importance, we investigated the relationships between these concepts, thereby enhancing our understanding of the semantic structure of the data.

The content analysis encompassed the titles and abstracts of all 107 articles. It revealed five prominent themes: “public”, “policy”, “factors”, “governance”, and “culture”, which collectively comprised 35 distinct concepts. Figure 2 depicts the thematic and conceptual

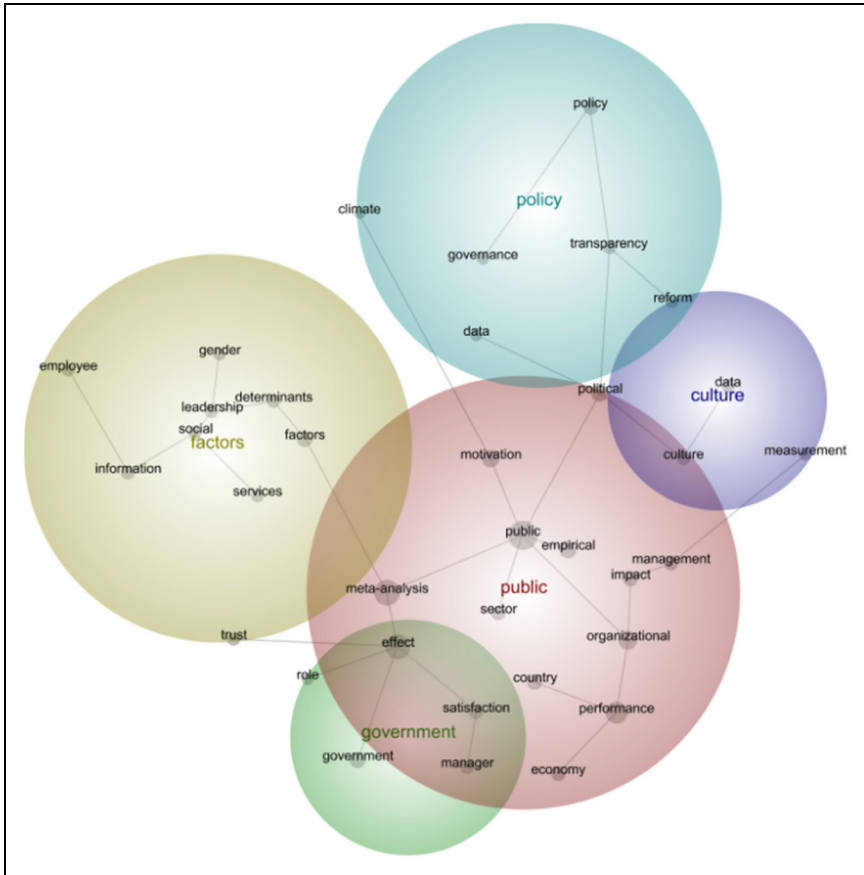


Figure 2. Theme and concept map of meta-analysis in public administration ($n = 107$).

landscape derived from the dataset. This three-dimensional (3D) visualization can discern the interconnections and associations among the 35 concepts. Notably, the theme of “public” emerges as predominant, followed by “factors”, indicating that meta-analyses in the Public Administration domain typically investigate prevalent antecedents and consequences of specific topics. Within the “policy” theme, significant attention is directed towards “performance” (94 mentions) and “organization” (60 mentions). Moreover, the map highlights several moderator factors, such as data type (34 mentions), culture (29 mentions), and measurement (23 mentions).

The interconnected themes of “public” (375 occurrences), “policy” (235 occurrences), and “governance” (163 occurrences) underscore the emphasis of meta-analyses in the public administration domain on *public policy* and *public governance* matters. Figure 2 illustrates that research topics span both micro- and macro-levels. *Macro-level* considerations encompass aspects like “public performance” (97 mentions), “public service” (66 mentions), “public management” (52 mentions), “climate change”

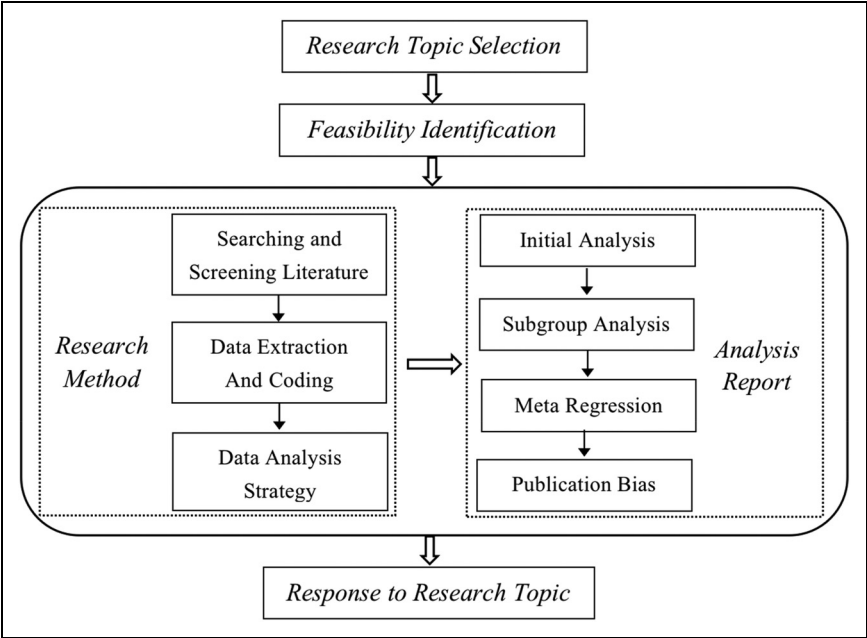


Figure 3. An analytical framework for a meta-analytic research report.
Note: Adapted from Ringquist (2013).

(37 mentions), and “political policy” (29 mentions). Conversely, *micro-level* topics include “*trust*” (37 mentions), “*satisfaction*” (32 mentions), and “*motivation*” (31 mentions).

Literature review based on the framework of meta-analysis

Through a review of 107 meta-analytic studies, our study has developed a framework for analyzing and assessing meta-analytic reports (see Figure 3). The meta-analysis process is delineated into five principal segments: selection of research topics, identification of feasibility, delineation of analytical methods, presentation of analytical findings, and addressing the research questions.

This research conducted an exhaustive review of meta-analytic studies published in five prominent journals: *International Review of Administrative Sciences (IRAS)*, *Journal of Public Administration Research and Theory (JPART)*, *Public Administration (PA)*, *Public Administration Review (PAR)*, and *Public Management Review (PMR)*. These journals are recognized for their ability to reflect cutting-edge developments in public administration. Through systematic screening in the Web of Science database and manual searches on these journals’ official websites, a total of 34 meta-analytic articles published within the last decade (2015–2024) in these journals were identified for analysis.

Building upon the analytical framework proposed for meta-analytic studies in public administration, we meticulously documented and synthesized the meta-analytic procedures employed in the 34 selected articles in the five prominent journals, elucidating the key steps inherent in meta-analysis within both methodology and result analysis.

Research topic selection

Determining the research theme and focusing on research questions are the critical initial steps in conducting meta-analysis. Upon reviewing 34 articles, it was found that the exploration of the relationship between independent and dependent variables in existing meta-analyses falls into three main categories: single independent variable with the single dependent variable (e.g. Park, 2020), single independent variable with multiple dependent variables (e.g. Backhaus and Vogel, 2022), and numerous independent variables with the single dependent variable (e.g. Harari et al., 2016). Among these, the primary research focus of current meta-analyses remains on investigating the relationship between two specific variables (one-to-one), comprising 17 articles. Additionally, there are eight meta-analyses focusing on core variables for particular topics, exploring factors influencing these variables (many-to-one). The selection of meta-analysis research topics is based on divergent conclusions existing in previous empirical studies. If existing research findings are consistent or similar, the necessity of conducting a meta-analysis is questioned. Therefore, the topics selected for meta-analysis originate from the “conflicting” conclusions in existing studies, which pose a “puzzle” for research. The problem statements in the 34 articles address the “conflicts” existing in previous literature. Topic analysis reveals that the topics mainly focus on individual-level motivations and attitudes and organizational-level performance, including public service motivations, trust, satisfaction, and organizational performance.

Feasibility identification

Feasibility analysis elucidates the necessity of conducting meta-analysis research. Summarizing the discussions in this aspect across the 34 articles, it can be categorized into “one premise and two pathways”. The premise is that the focused topic is a widely discussed hot topic, with a rich accumulation of empirical studies in the preliminary stage. The two pathways comprise the following: (1) existing empirical studies show disputes when exploring the influence of relevant variables, which is most common in “one-to-one” meta-analysis, for instance, Homberg et al. (2015) stating in their meta-analysis on public service motivation and satisfaction that existing studies present controversial conclusions that urgently need to be explored; and (2) systematic reviews are conducted on a specific hot topic, focusing on a theoretical framework or core concepts for meta-analysis. For example, Zhang et al. (2022) focused on the expectancy disconfirmation model and systematically analyzed various pathways of the model. Feasibility analysis also involves deducting theoretical foundations and research hypotheses. As meta-analysis is a literature study conducted with empirical quantitative thinking, it aligns with the empiricist path, including proposing hypotheses, establishing theoretical models, and validating them with empirical data (Lee and Hung, 2022).

Among the included literature, a total of 26 articles deduced research hypotheses based on existing theories.

Research method

Searching and screening literature. Sufficient empirical literature forms the foundation and starting point for conducting meta-analysis. Existing meta-analyses demonstrate a combination of uniformity and diversity in database selection. Uniformity is evident in using two major search databases—Web of Science and Google Scholar—with 21 articles opting for Web of Science and 16 articles choosing Google Scholar. Diversity is reflected in the utilization of various databases such as EBSCO, ProQuest, and JSTOR, or the exclusive selection of SSCI journals in public administration for retrieval, along with supplementary searches of references cited in the included literature. Additionally, some studies explicitly report searches and inclusions of working papers, including using platforms like the Social Science Research Network (SSRN), searching for conference papers, and reaching out to authors in the field. However, only eight articles explicitly reported the search and screening of working papers, accounting for only 23.5%.

Reporting the criteria and procedures for literature search and screening is a crucial step in meta-analysis, with 25 articles providing explanations of varying lengths. Of these, 22 articles utilized Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) to document the screening process, accounting for 64.7%. Additionally, we conducted statistical analysis on the number of included articles and effect sizes. Among the 34 meta-analyses, the maximum number of included articles was 175 (Humphrey, 2023), while the minimum was 17 (Zhang et al., 2022), with an average of 54 included articles. Regarding effect sizes, the maximum was 2188 (Gerrish, 2016), the minimum was 20 (Eshuis et al., 2021), and the average included effect sizes amounted to 247. It is evident that there is no strict limitation on the number of included articles and effect sizes. According to the 34 studies, there should be at least three effect sizes per subgroup analysis to ensure the reliability of the analysis. Power calculations for subgroup analyses should be performed because Cuijpers et al. (2021) demonstrated that subgroup analyses typically require 3–4 times as many studies as the main analysis to ensure adequate power.

Data extraction and coding. Data extraction and coding from literature are crucial for ensuring the quality and replicability of meta-analysis. Through a systematic review of 34 articles, it was found that detailed descriptions of data coding and extraction were lacking in five articles. Data coding involves recording basic data and potential moderator variables, with a more critical aspect being the standardization of effect sizes reported by individual studies (Wang and Guan, 2022). Among these, 20 articles presented the coding and extraction of basic data in table format within the main text or appendices, providing a clearer presentation of the fundamental data from each study. Effect size calculation bases were reported in textual form in all included articles, comprising three types: correlation coefficient (r), Cohen's d , odds ratio (OR), and standardized mean difference (SMD). In the process of coding conversion, 21 articles transformed various values (including t -values, β -values, OR-values, Cohen's d -values) into correlation coefficient

(*r*)-based values using formulae. However, seven articles did not provide detailed explanations of the coding conversion and extraction process. This study summarized the most widely used formulae for converting *r*-values (see Appendix, Table 1).

Standardization and accuracy of coding conversion are foundational for ensuring the scientific reliability of subsequent analysis results. Among the 34 meta-analyses, none conducted evaluations of the selected literature based on correlation indicators. Although the criteria for literature data selection were reported, evaluation indicators were not provided, indicating a lack of systematic evaluation of the original literature. Some studies calculated reliability based on coding consistency, with the calculation standard limited to the consistency coefficient of coding personnel data extraction, without evaluating the quality of the articles themselves. Compared with the more systematic evaluation standards in medical and psychological fields (such as the Agency for Healthcare Research and Quality, Newcastle Ottawa Scale), the public administration field requires the construction of relevant indicators.

Data analysis strategy. The widespread interdisciplinary application of meta-analysis provides various pathways for its analytical strategies. The “meta” and “metafor” analysis packages based on R software are the most commonly used analytical tools. Additionally, user-friendly software such as Comprehensive Meta-Analysis (CMA), Stata, Meta-Essential, and Dmetar packages are also utilized. In terms of analytical strategies, some studies reported using sample size-weighted averages as the mean effect size for handling multiple effect sizes, with 26 articles employing random-effects models for data analysis. Upon reviewing the details of data analysis, only seven studies provided detailed reports of their analytical strategies in the appendices.

Analysis report

Initial analysis. The preliminary synthesis analysis is crucial in meta-analysis as it focuses on resolving existing controversial conclusions. This preliminary analysis involves calculating weighted average effect sizes from original studies, aiming to provide a unified answer to the primary question addressed in meta-analysis, which is to reconcile conflicting research conclusions (Backhaus and, Vogel, 2022; Gerrish, 2016). Generally, preliminary analysis involves two different hypotheses: fixed effects and random effects. The fixed effects hypothesis assumes that the included studies share a common true effect size, and each study provides a valid estimate of this effect, with errors stemming from sample size; while the random effects hypothesis posits that the included studies do not share a common true effect size, but rather exhibit random effect sizes following a normal distribution. Selection between fixed effects or random effects can be determined through heterogeneity tests using the Chi-square and I^2 statistics (Fan et al., 2022; Field and Gillett, 2010). In social science research, studies typically align more with the random effects hypothesis (Humphrey, 2023; Ringquist, 2013). When the *Q*-value in the heterogeneity test exceeds the degrees of freedom, the *p*-value is less than 0.05, and I^2 is greater than 75%, it indicates heterogeneity among included studies, necessitating the use of random effects meta-analysis for

calculation. Among the meta-analyses included, 23 studies explicitly reported the use of a random effects meta-analysis (e.g. Chen and Ganapati, 2023).

The measurement standards for effect size magnitude in preliminary analysis generally follow the distinction made by Cohen (1992). According to this standard, for the effect size of correlation coefficient (r), 0.10 is considered a small effect size, 0.3 is moderate, and 0.50 is large. Lipsey and Wilson (2001) suggest 0.10 for a small effect size, 0.25 for moderate, and 0.30 for large. Although specific numerical values of effect sizes were reported in all 34 meta-analyses, only six studies provided interpretations of effect sizes based on these two standards. Additionally, forest plots effectively display effect sizes and their confidence interval estimates from included studies, allowing for trend analysis of effect size changes over time based on publication year sorting (Pigott and Polanin, 2020). Forest plots also clearly identify extreme values of included effect sizes, providing a basis for subsequent robustness testing or publication bias analysis. Among the 34 meta-analyses, 12 studies utilized forest plots to present included effect sizes.

Subgroup analysis. Subgroup analysis is conducted on the basis of preliminary analysis to identify the reasons causing differences in effect sizes among various subgroups of analysis variables. As a statistical analysis method, subgroup analysis allows scholars to divide study subjects into different subgroups based on characteristics such as culture, nationality (institutional contexts, policy implementation styles, and governance systems across countries), group traits, measurement methods, and variable categories in meta-analysis (Parola et al., 2019; Voorn, 2021). Effect sizes are then estimated for each subgroup and compared between subgroups to study interactions or effect modification, determining whether the effect sizes vary under different traits or conditions. Among the 34 meta-analyses reviewed, 23 studies conducted subgroup analyses based explicitly on national context, cultural background, or institutional tradition—dimensions that are central to comparative public administration. Notably, nearly half of these cross-national comparisons simply divided cases into US versus non-US contexts (George et al., 2019; Wang and Guan, 2022), limiting the scope of comparative insight. These studies often remain restricted to a small set of countries or administrative contexts and rarely provide systematic cross-national comparisons.

Meta regression. Meta-regression analysis models the relationship between effect size and covariates by introducing one or more predictors (Wang and Guan, 2022). Compared with subgroup analysis, meta-regression analysis can comprehensively consider variables from different groups, providing a deeper exploration of research results' heterogeneity (Humphrey, 2023). Meta-regression allows exploration of the effects of different covariates on effect size, offering more explanatory power in the analysis. Therefore, a preferable approach is to use subgroup analysis to identify categorical variables causing differences in included study effect sizes and then incorporate significant subgroup variables into the model through meta-regression for a more explanatory moderation analysis (Fan et al., 2022; Homberg et al., 2015). Among the 34 meta-analyses included, 19 studies utilized meta-regression analysis for testing covariates, while 13 studies used subgroup analysis and subsequently conducted meta-regression. While meta-regression

allows incorporation of multiple covariates, including national or cultural characteristics, most existing public administration meta-analyses have only analyzed specific cultural dimensions, such as power distance (Zhang et al., 2022) or collectivism (Humphrey, 2023), and have not fully exploited this potential to conduct comprehensive cross-national comparisons. This gap highlights the need for systematic, comparative public administration-focused meta-analytic approaches.

Publication bias test. Publication bias is often criticized in meta-analyses of journal articles, suggesting that findings with statistical significance are more likely to be published, while null results are less likely to be published (Rosenthal, 1979). Consequently, there may be bias in the statistical synthesis of published articles. To identify potential publication bias issues, two methods—namely, visual and statistical testing—were employed. Specifically, following the recommendations of Ringquist (2013) and Field and Gillett (2010), visual testing for publication bias was conducted by presenting a symmetrical funnel plot, along with Egger's test and Begg's test for statistical testing to detect publication bias. For funnel plots, if there is no publication bias in the included original studies, the dots representing each effect size should be evenly and symmetrically distributed on both sides of the average effect size and exhibit a funnel-shaped distribution. The position of the dots can also serve as a basis for analysis: the closer the dots are to the bottom of the funnel, the higher the precision. Additionally, funnel plots can be used to detect and identify outlier study results, providing a basis for subsequent sensitivity analysis. Egger's test and Begg's test, on the other hand, can be used to discriminate based on significance coefficients.

It is important to emphasize that the existence of publication bias is not a criterion for judging the objectivity of a meta-analysis study (George et al., 2019). Rather, the focus should be on accurately reporting and explaining the reasons for the existence of publication bias. Some studies used sensitivity analysis to detect publication bias (Parola et al., 2019). All 34 studies reported on the presence of publication bias, with Egger's test and funnel plots being the most commonly used indicators.

Response to research topic

The discussion section including the response to research topic is crucial for meta-analysis. Meta-analysis allows us to carefully analyze and combine results from various studies in a systematic way. This approach boosts the overall reliability and applicability of research findings, providing valuable insights that are useful for policy-makers, professionals, and scholars alike (Park, 2020; Parola et al., 2019). To response the research topic, scholars interpret the findings of their study, provide context for their results, discuss the implications, and offer insights into future research directions. All 34 meta-analysis research presented the interpretation of findings and explored both the theory contributions and implications for practice. While this section does not adhere to stringent formatting guidelines, it remains imperative to elucidate the findings of the meta-analysis, delineate its theoretical and practical implications, and suggest avenues for future research.

Recommendations for meta-analysis in public administration

Meta-analysis, as a quantitative systematic literature review method, has been widely used in disciplines such as biomedical science, psychology, and organizational behavior, providing a valuable reference for optimizing the processes for meta-analysis in public administration. This article offers four suggestions aimed at providing guidance for the optimization of meta-analysis in public administration in the age of open science.

Preregistration for enhanced transparency and rigor

Preregistration is a tool for improving research transparency, credibility, and promoting research standardization, scientific rigor, academic supervision, and contributing to the healthy development of academic science (Evans et al., 2023; Logg and Dorison, 2021). Preregistration in meta-analysis research refers to professionals clearly and registering the design, analysis plan, and hypotheses of the study before conducting the meta-analysis, and then publicly registering this information on a specialized platform (Moreau and Gamble, 2022). Preregistration requires researchers to publicly document their study design, hypotheses, inclusion criteria, and analysis plans before data collection, mitigating risks like *p*-hacking and selective study inclusion. Preregistration is widely used in fields such as medicine and psychology (Adam, 2019; Moreau and Gamble, 2022). Currently, experimental research in public administration is actively promoting and using preregistration, such as field experiments in public administration (Yang et al., 2023). However, the use of preregistration remains rare in meta-analyses within the field of public administration. There is PROSPERO preregistration in the field of health medicine (<https://www.crd.york.ac.uk/prospero/>) (Fan et al., 2021), and OSF (Open Science Framework) preregistration for meta-analysis in psychology (<https://osf.io/>) (Parry et al., 2021). The Institute of Psychology of the Chinese Academy of Sciences has established a psychological preregistration platform (<https://os.psych.ac.cn/preregister>), providing a reference for the standardization of academic journal publications in psychology. Therefore, in order to construct the development of disciplines under an autonomous knowledge system, it is necessary to establish a meta-analysis preregistration system suitable for the field of public administration based on scientific research institutions or journal platforms, standardize the development of the discipline, and improve the transparency and standardization of meta-analysis in public administration.

In drawing from other disciplines' preregistration systems, the necessary information for the preregistration system of meta-analysis in the field of public administration should include the following essential details: research topic, keywords, literature search date, timeline, rationale for focusing on the topic, search terms, literature search steps, literature coding records, data analysis strategy, basic information of the research team, and funding sources. Submission of preregistration records should be required for meta-analysis research submissions to enhance the scientific rigor of the studies.

Ensuring quality with self-check reporting standards

The meta-analysis research self-check report is crucial for enhancing quality. It involves scholars conducting a self-examination and reporting process on their meta-analysis study upon completion (Meyer, 2018). This helps ensure the transparency and reproducibility of the research and aids in identifying potential methodological or analytical issues (Moreau and Gamble, 2022). Currently, some leading journals in psychology have explicitly outlined the standards for meta-analysis research self-check reports in their manuscript submissions (e.g. *Psychological Science*, *Acta Psychologica Sinica*). The Agency for Healthcare Research and Quality Checklist (Flottorp et al., 2013) for the cross-sectional studies assessment and the Newcastle Ottawa Scale (Stang, 2010) for longitudinal studies assessment are popular tools in medical meta-analysis research self-check reports. Journals in the field of public administration should establish review and assessment criteria for meta-analysis submissions, setting disciplinary standards and making the self-check report a mandatory option for submitting meta-analysis research manuscripts.

The self-check report for meta-analysis research in public administration should include the necessity and theoretical contribution of the meta-analysis, literature search and data coding processes, assessment of literature quality and publication bias, method of effect size calculation, registration website and preregistration ID, and similar topic publications (if they exist). Submitting a self-check report in a meta-analysis enables editors and peer reviewers to quickly understand the research topic, framework, and operating procedures. Including the self-check report in the article's Appendix allows readers to comprehend the theoretical and methodological aspects of the study. Therefore, there is an urgent need in the field of public administration to create a meta-analysis self-check report tailored to the discipline, which will not only enhance research standardization but also further improve the readability of the research.

Advancing theory-driven meta-analysis in public administration

Theoretical contribution serves as the cornerstone of meta-analysis research. Constructing meta-analyses grounded in theoretical frameworks is essential to enhance the explanatory power and relevance of public administration research, particularly in comparative public administration, where variations in national institutions, administrative traditions, and cultural factors shape governance outcomes. Our content analysis of 107 meta-analytic articles reveals that 73.5% focus on correlations between specific variables (e.g. trust and performance), limiting their theoretical depth. This overemphasis on variable relationships, rather than dissecting theoretical models, underscores the need for *theory-driven* meta-analysis in public administration. By aligning meta-analysis with theoretical frameworks, professionals can not only provide a comprehensive understanding of the phenomenon under study but also contribute to theory development and refinement (Zhang et al., 2022).

However, in the field of public administration, meta-analyses predominantly focus on moderator variable analyses (including subgroup analysis or meta-regression analysis),

with limited exploration of mediation mechanisms. It is imperative to introduce cross-temporal meta-analysis (Karazsia et al., 2017; Rudolph et al., 2020) and structural equation model meta-analysis (Cheung, 2015, 2019) into the realm of public administration. Additionally, it is advisable to explore the construction of a three-level meta-analysis. Considering meta-analysis from a two-level perspective (original studies–effect sizes) often leads to an overestimation of the overall effect owing to the presence of multiple effect sizes within individual studies (Lipsey and Wilson, 2001). A three-level meta-analysis, however, offers a more effective solution to address the dependency issue arising from multiple effect sizes within the same study. By retaining the original data, three-level meta-analysis enhances statistical power and theoretical contribution (Assink and Wibbelink, 2016). To implement theory-driven meta-analysis, public administration researchers should align research questions with established theories, specify theoretical moderators (e.g. cultural or institutional factors), and interpret findings in the context of these frameworks. This approach not only addresses the current methodological gaps identified in our systematic review of 34 articles from leading journals (2015–2024) but also strengthens the applicability of meta-analytic findings for evidence-based policymaking in cross-national and comparative public administration contexts, highlighting how meta-analysis can uncover systematic patterns across diverse governance systems.

Enhancing insights with mixed methods meta-analysis

Mixed methods involve the integration of different research approaches to provide a comprehensive understanding of a research problem or phenomenon (Harrits and Møller, 2021). In the context of combining experimental research with meta-analysis, or repeated experiments with meta-analysis, scholars can leverage the strengths of both approaches to enhance the validity and generalizability of their findings (Eshuis et al., 2021; Weibel et al., 2009). This integration is particularly valuable for comparative public administration studies, where cross-national experiments or replicated interventions can be systematically synthesized to understand context-dependent effects.

The experimental plus meta-analysis approach represents a synergistic blend of the methodological strengths of experimental research and the analytical power of meta-analysis. In this approach, researchers initiate investigations through controlled experiments, meticulously designed to ascertain causal relationships or to test specific hypotheses within tightly controlled conditions. Following the experimentation, researchers employ meta-analysis techniques to systematically collate and analyze data from multiple empirical studies addressing the same research question. By explicitly considering country- or culture-specific factors, this approach can generate insights relevant to comparative public administration, highlighting how institutional or administrative differences shape observed outcomes. This process facilitates the synthesis of findings across diverse independent studies, thereby offering a more comprehensive evaluation of the overall effect size and unveiling underlying trends or moderators.

In the context of the replication plus meta-analysis approach, researchers undertake a series of experiments focusing on identical or analogous research inquiries, deliberately varying certain parameters or contexts across individual studies. Post experimentation,

meta-analysis is applied to amalgamate the outcomes of these replicated experiments, thereby enabling a meticulous examination of consistency and effect magnitude across distinct settings or conditions. In comparative public administration, such replication across countries or administrative contexts allows researchers to systematically test whether theoretical mechanisms are robust across different institutional environments. This methodological framework empowers researchers to assess the robustness of their findings and discern factors that potentially influence or moderate the effects observed in individual experiments (Eshuis et al., 2021).

By incorporating meta-analysis into experimental or replicated experimental designs, researchers enhance the empirical foundation of their investigations, thereby fostering more dependable conclusions regarding the phenomena under scrutiny. Moreover, the integration of mixed methods not only enriches the methodological repertoire but also facilitates a more holistic and nuanced comprehension of intricate issues within the domain of public administration, especially in comparative and cross-national contexts (McDonald III et al., 2022; Mele and Belardinelli, 2018).

Conclusion

Meta-analysis, as a comprehensive review method rooted in quantitative research, seeks to explore the connections between fundamental concepts and assess theoretical models by reevaluating existing quantitative studies. This study presents a comprehensive analysis of 107 meta-analyses within the field of public administration, resulting in the development of a structured framework for conducting such analyses. Through a systematic review, we have delineated key elements comprising topic selection, feasibility assessment, research methodology, and reporting procedures. By applying this framework to 34 articles from leading journals, we have effectively outlined and categorized the procedural steps involved in meta-analysis. Considering the current landscape of meta-analysis, our study advances four interdisciplinary recommendations. These recommendations aim to bolster optimization and theoretical robustness within the practice of meta-analysis, thus fostering further progress in research within the domain of public management. By implementing these recommendations, researchers can enhance the credibility and reliability of meta-analytical findings, contributing to the ongoing advancement of knowledge in public administration.

Certainly, it is important to acknowledge that any research methodology evolves along a path of continual refinement. The interdisciplinary recommendations proposed in this study are not static but subject to updates and revisions as the field progresses. As new insights emerge and methodologies evolve, it is imperative for scholars to remain open to adaptation and refinement of their approaches. Therefore, the recommendations provided in this study offer a novel framework for enhancing the standardization and theoretical robustness of meta-analysis within the realm of public administration. However, they should be viewed as dynamic guidelines that may undergo modifications in response to emerging research trends and advancements in the field (McDonald III et al., 2022). By embracing this dynamic nature of scholarship, researchers can effectively navigate the evolving landscape of meta-analysis and contribute meaningfully to the ongoing development of knowledge in public administration in the age of open science.

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Supplemental material

Supplemental material for this article is available online.

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